CISCO PACKET TRACER

Enterprise Network Simulation Projects

Prepared by

Ath Vandy

Field

Information Technology / Computer Networking

Tools Used

Cisco Packet Tracer • Cisco IOS CLI • GitHub

Table of Contents

Chapter 1: Introduction

This project portfolio demonstrates practical networking skills using Cisco Packet Tracer. The purpose of these simulations is to design and configure enterprise network environments using Cisco routers and switches. The projects focus on VLAN implementation, inter-VLAN routing, DHCP services, and routing protocols including RIP, OSPF, and BGP.

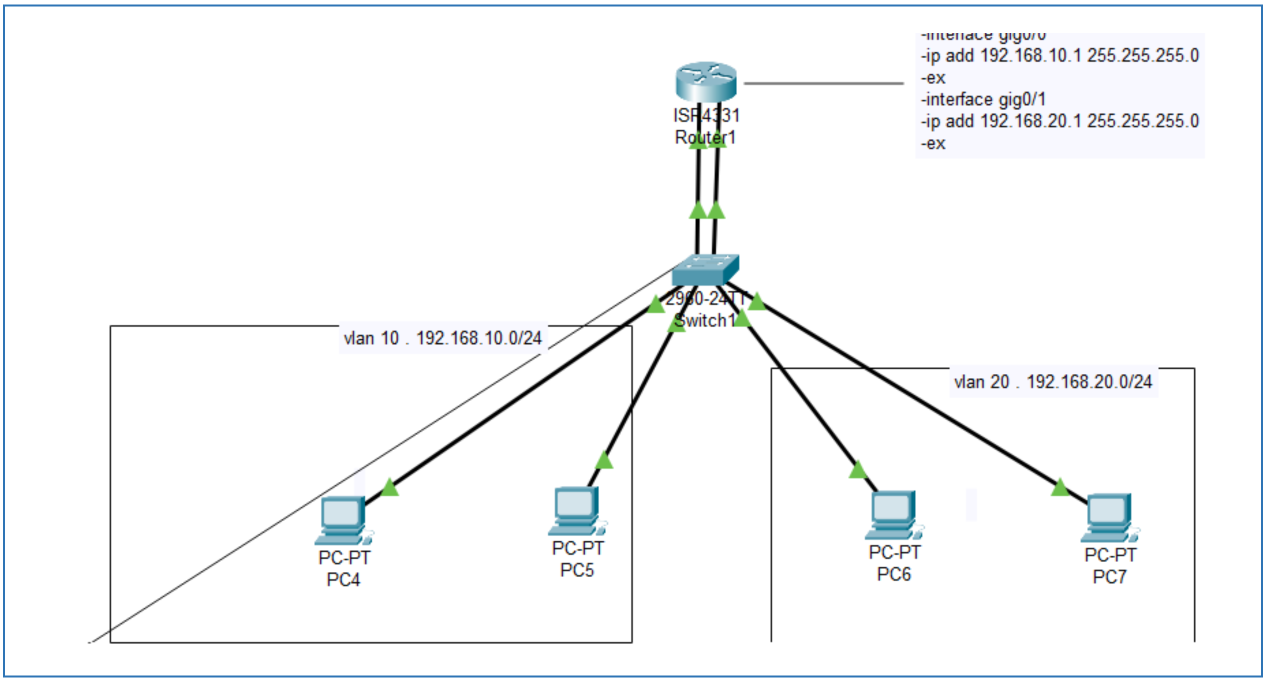
Each project was built and tested inside Cisco Packet Tracer, with configurations verified using standard Cisco IOS CLI commands. The report documents the design rationale, the exact configuration steps taken, and the verification output confirming that each network behaved as intended. Where issues arose during testing, they are documented alongside the troubleshooting process used to resolve them.

1.1 Project Objectives

- Design and simulate small-to-medium enterprise network topologies
- Configure VLANs to logically segment traffic by department or function
- Implement inter-VLAN routing using the router-on-a-stick method
- Deploy DHCP services for automatic IP address assignment
- Configure and compare static routing with dynamic routing protocols (RIP, OSPF, BGP)
- Verify network functionality using Cisco IOS show commands and connectivity tests
- Diagnose and resolve common configuration issues

Chapter 2: Network Design Overview

The overall architecture used across these projects follows a typical small-enterprise design: a single edge router connects to the outside network (representing the Internet or an upstream ISP), while two access-layer switches handle traffic for distinct VLANs — one for end-user workstations and one for servers.



2.1 Devices Used

Device	Role	Quantity
1x Cisco Router (2911 series)	Inter-VLAN routing / default gateway	1
2x Cisco Layer 2 Switch (2960 series)	VLAN segmentation and access-layer switching	2
PCs / End Devices	Represent workstations and servers	4+
Serial / Copper links	WAN and LAN interconnects	As needed

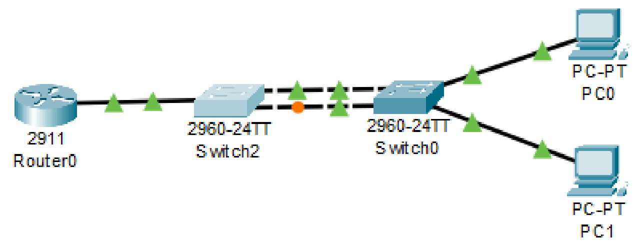
2.2 Communication Flow

- End devices in VLAN 10 and VLAN 20 send traffic to their local switch
- Inter-VLAN traffic is forwarded via trunk links to the router
- The router performs Layer 3 routing between VLAN subnets
- Outbound traffic is routed toward the simulated Internet edge

Chapter 3: VLAN Using Trunk Mode

3.1 Network Topology

Topology used for VLAN and trunk configuration:



3.2 Configuration

The following commands were applied on the switches to create the VLANs and configure the trunk link:

```
enable
configure terminal

vlan 10
name SALES

vlan 20
name IT

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/24
switchport mode trunk
```

3.3 Verification

Verification was performed using the following commands:

```
show vlan brief

show interfaces trunk
```

Result:

✔ VLAN created successfully

✔ Trunk communication established

Chapter 4: Router-on-a-Stick

4.1 Topology and Addressing

Router subinterfaces were used to route between VLANs over a single trunked link to the switch.

IP Address Table

Device	Interface	IP Address
Router	G0/0.10	192.168.10.1
Router	G0/0.20	192.168.20.1
PC1	DHCP	192.168.10.x

4.2 Configuration

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

4.3 Verification

```
show ip interface brief
ping 192.168.20.1
```

✓ Subinterfaces active and routing between VLAN 10 and VLAN 20

Chapter 5: VLAN with DHCP

DHCP was configured on the router to automatically assign IP addresses to end devices, removing the need for manual static configuration on every PC. This is standard practice in enterprise networks, where dozens or hundreds of hosts would otherwise require individual addressing.

5.1 DHCP Pool Configuration

```
ip dhcp pool VLAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

5.2 Testing

```
ipconfig
ping gateway
```

✓ PCs received IP addresses automatically from the DHCP pool

✓ Gateway reachable from all DHCP clients

Chapter 6: Routing Protocols

This chapter compares static routing with three dynamic routing protocols — RIP, OSPF, and BGP — each configured and verified in a separate simulation.

6.1 Static Routing

Static routes are manually configured and best suited to small, stable networks where routes rarely change.

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

6.2 RIP (Routing Information Protocol)

```
router rip
version 2
network 192.168.1.0
network 10.0.0.0
```

Verification:

```
show ip route
show ip protocols
```

6.3 OSPF (Open Shortest Path First)

```
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
```

Verification:

```
show ip ospf neighbor
```

6.4 BGP (Border Gateway Protocol)

BGP is used to exchange routing information between different Autonomous Systems and is the routing protocol underlying the global Internet.

```
router bgp 65001
neighbor 10.0.0.2 remote-as 65002
network 192.168.1.0 mask 255.255.255.0
```

Verification:

```
show ip bgp summary
```

6.5 Protocol Comparison

Protocol	Type	Best Suited For
Static Route	Manual	Small, stable networks
RIP v2	Distance-vector, dynamic	Small networks, simple topology
OSPF	Link-state, dynamic	Medium-large enterprise networks
BGP	Path-vector, inter-AS	Internet-scale, multi-AS routing

Chapter 7: Troubleshooting and Testing

Documenting real troubleshooting steps demonstrates practical problem-solving ability, not just configuration copying.

7.1 Problem 1 — VLANs Cannot Communicate

Symptom: PCs in different VLANs, or on different switches, cannot ping one another

Cause: Incorrect trunk configuration between switches

Solution: verified the trunk state, then corrected the port mode.

```
show interfaces trunk

! corrective action:
switchport mode trunk
```

✓ Trunk re-established; inter-VLAN traffic restored

7.2 Problem 2 — OSPF Neighbors Not Forming

Symptom: Router does not see expected OSPF neighbors

Cause: Incorrect network statement / wildcard mask in OSPF configuration

Solution: verified neighbor state, then corrected the OSPF network statement to match the actual subnet.

```
show ip ospf neighbor
```

✓ OSPF adjacency formed successfully after correction

Chapter 8: Skills Developed

- Cisco IOS command-line configuration
- Network topology design
- VLAN segmentation
- Switching configuration (access and trunk ports)
- Routing protocol implementation (Static, RIP, OSPF, BGP)
- IP addressing and subnetting
- DHCP configuration and service deployment
- Network troubleshooting and root-cause analysis
- Connectivity testing and verification
- Enterprise network simulation using Cisco Packet Tracer

Conclusion

These projects demonstrate my ability to design and configure small enterprise networks using Cisco technologies. Across each simulation, I built the topology, applied the appropriate Layer 2 and Layer 3 configurations, and verified functionality using standard Cisco IOS diagnostic commands. Where problems occurred, I documented the troubleshooting process from symptom to root cause to resolution.

The experience gained from these simulations provides a strong foundation for professional roles such as Network Engineer, Data Center Engineer, Cloud Network Engineer, and IT Support Engineer. I look forward to applying and expanding these skills in a professional networking environment.